

Nilson Cardoso

Computer Science Graduate · Backend & Data

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PROFILE

Computer Science graduate with hands-on experience in backend development, REST API integration, and relational databases (PostgreSQL, MySQL). Built end-to-end data pipelines covering ingestion, cleaning, feature engineering, and prediction, and gained exposure to mainframe systems (COBOL, JCL, VSAM, z/OS) through a specialized training programme. Comfortable working across the full development cycle, from requirements gathering to testing and documentation. Looking to contribute to backend engineering and data-driven projects, applying Java, Python, SQL, and API development skills to build reliable, scalable solutions.

PROJECTS

RiskTrack | Final Year Project | 2025 | Grade: 18/20

End-to-end web application for student dropout risk prediction, built with R/Shiny and PostgreSQL, covering the full data workflow from ingestion to prediction within a single integrated pipeline.

- Designed and implemented a data pipeline: raw CSV upload → cleaning/validation → feature engineering → model training (decision tree, CRISP-DM) → prediction output
- Defined functional and non-functional requirements and documented use cases covering data upload, model training, and prediction flows
- Performed functional testing across core features, ensuring platform stability and data validation
- Used PostgreSQL for structured data storage and R/Shiny for the interactive web interface

TravelEx | Team Project · Agile/Scrum | 2025 | Grade: 18/20

Peer-to-peer package delivery platform for Angolan and international travelers, developed collaboratively across multiple sprints.

- Developed the backend with Node.js + TypeScript, building and integrating REST APIs consumed by the frontend
- Integrated Firebase Firestore (NoSQL database) and Firebase Auth for user authentication and data persistence
- Contributed to requirements definition, sprint planning, and documentation across multiple sprints
- Participated in functional testing and API validation throughout the development cycle

SKILLS

Languages: Java, Python, R, SQL, JavaScript, TypeScript, COBOL, HTML, CSS

Databases: PostgreSQL, MySQL, Firebase Firestore (NoSQL), DB2

Backend & Integration: REST API integration and development, Node.js, microservices concepts, Firebase Auth

Data & Analytics: Data transformation, ETL workflows, data validation, Power BI, Excel

Dev Tools & Practices: Git, Docker (familiar), CI/CD (familiar), Agile/Scrum, requirements analysis, functional testing

Mainframe (Additional Training): COBOL, z/OS, JCL, VSAM, ISPF/TSO, HOGAN

TRAINING & CERTIFICATIONS

COBOL & HOGAN Academy — Sinensia / DXC Technology (2025)

EDUCATION

Bachelor of Computer Science · Universidade Lusófona do Porto · 2025

LANGUAGES

Portuguese: Native | English: B2